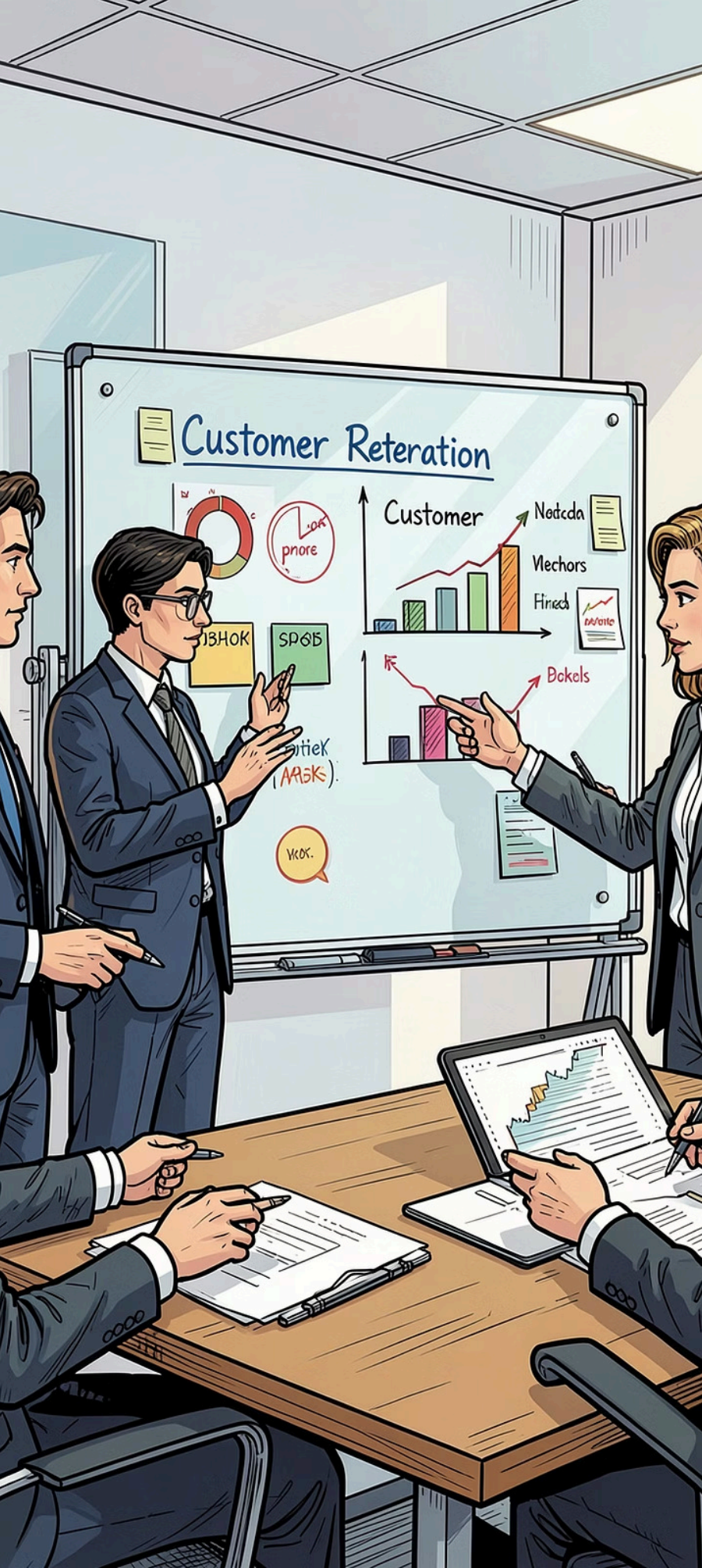




Customer Churn Analysis

Identifying attrition drivers and delivering actionable retention strategies to protect revenue.

PREPARED BY: RADHIKA VISHWAKARMA · IBM



Executive Summary

7K

Total Customers

Records in the Telco dataset

26.54%

Observed Churn Rate

Based on the source dataset

\$139K

Monthly Revenue Lost

From churned customers

\$1.7M

Annual Revenue at Risk

Total potential impact

Key finding: Month-to-month contracts, certain payment methods, and customers with fewer subscribed services are most likely to churn.

Project Objectives



Measure Churn & Impact

Quantify customer attrition and its direct business impact.



Identify High-Risk Segments

Pinpoint customer groups with the highest churn risk.



Analyze Key Variables

Contract type, tenure, payment method, internet service, and demographics.



Actionable Recommendations

Convert insights into retention strategies and business actions.



Power BI Dashboard

Interactive visualizations and KPI reporting for decision-makers.

Technologies Used

Python

Data cleaning, transformation, and exploratory analysis

Pandas & NumPy

Data manipulation, aggregation, and numerical operations

Matplotlib & Seaborn

Visualizations and trend analysis

SQL

Data extraction, filtering, aggregation, and business queries (~137 lines)

Power BI

Interactive dashboard development and KPI reporting

Methodology

01

Data Collection

Imported the Telco Customer Churn dataset

03

Exploratory Data Analysis

Investigated churn behavior across customer segments

05

Dashboard Development

Built interactive Power BI visualizations and filters

02

Data Cleaning

Handled missing values, corrected data types, standardized fields

04

SQL Analysis

Executed business queries to identify patterns and KPIs

06

Recommendation Framework

Converted insights into business actions

Main Dashboard Overview



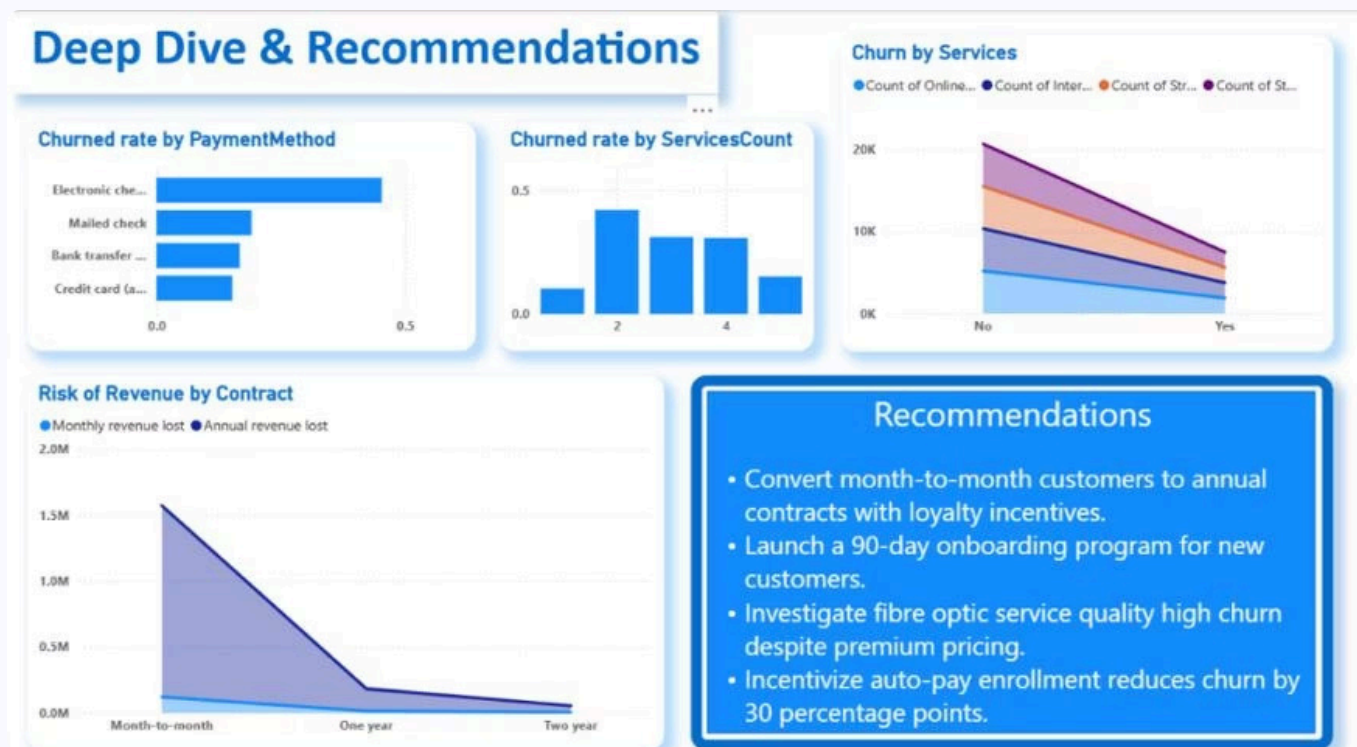
Dashboard Filters

- Contract: Month-to-month, One year, Two year
- Senior Citizen, Gender
- Internet Service: DSL, Fiber optic, No
- Payment Method

Key Metrics Tracked

- Churned rate by Internet Service (24.56% vs 69.4%)
- Churn by tenure, senior citizen, and contract type
- Sum of Monthly & Total Charges by payment mode

Deep Dive Analysis



Churn by Payment Method

Electronic check users show the highest churn rate, while auto-pay (credit card) users churn significantly less.

Churn by Services Count

Customers with more subscribed services are markedly more loyal – fewer services correlates with higher churn.

Revenue Risk by Contract

Monthly and annual revenue lost is concentrated in month-to-month contract holders.

Key Findings

Contract Type

Month-to-month contracts show the **highest churn risk**; longer-tenure customers are significantly more likely to stay.

Payment Method

Certain payment methods – especially electronic check – are associated with **higher churn**.

Services & Loyalty

Customers with **more subscribed services** tend to be more loyal and less likely to churn.

Senior Citizens

Senior citizens represent a **notable at-risk segment** requiring targeted retention focus.

Internet Service

Fiber optic customers show high churn despite premium pricing – **service quality concerns** may be a factor.

Business Recommendations



Loyalty Incentives

Convert month-to-month customers to annual contracts with loyalty incentives.



90-Day Onboarding

Launch a structured onboarding program for new customers to reduce early-stage churn.



Promote Auto-Pay

Incentivize auto-pay enrollment – reduces churn by up to **30 percentage points**.



Investigate Fiber Optic

Investigate fiber optic service quality – high churn despite premium pricing signals a potential issue.



Targeted Retention Campaigns

Develop retention campaigns for high-risk segments, including senior citizens.



Track KPIs Continuously

Monitor churn KPIs through the Power BI dashboard for ongoing decision-making.

Conclusion

This project combines **Python analytics**, **SQL-based business querying**, and **Power BI visualization** to deliver actionable customer-retention insights.

Understand Churn Drivers

Dashboard enables non-technical stakeholders to identify attrition factors

Reduce Revenue Loss

Informed decisions to improve customer retention and protect \$1.7M in annual revenue

Monitor KPIs

Track monthly and annual revenue at risk in real time

